

November 7th, 2025

Dear Editor,

We are submitting our manuscript, *High temperatures predict gut microbiome composition, metabolic capacity, and immune interactions in a wild primate species*, for review. Below are the answers to the required questions:

1. What scientific question is addressed in this manuscript?

This manuscript investigates how natural thermal variation affects gut microbiome composition and functional capacity in a wild non-human primate and whether these effects are related to host immunity.

2. What is/are the key finding(s) that answer this question?

We find that high temperatures are associated with changes in microbiome community-level diversity, taxonomic composition, and metabolic capacity consistent with physiological responses to heat stress. We also show that the microbiome's response to high temperatures is linked to host immunity and may affect gut-immune interactions.

3. Why is this work important and timely?

Microbiomes may play a key role in determining the thermal tolerances of their host species, highlighting an important pathway by which climate change could negatively affect species. However, this research has largely been conducted in laboratory settings and offers limited insight into how species respond to natural thermal variation.

4. Describe how your paper fits within the scope of GCB.

Our study addresses the concern that rising temperatures threaten wild animal populations, particularly in tropical species with narrow thermal tolerances. We provide rare evidence from a natural population of capuchins that changes to the gut microbiome at high temperatures may contribute to physiological heat stress by compromising essential microbiome functions.

5. What biological AND global change aspects does it address?

We evaluate changes to gut microbiome-immune interactions at high temperatures to address how rising temperatures due to climate change may impact wild animal populations through their microbiomes. While previous work in this area has largely focused on ectotherms, our study provides novel insight from a wild endotherm species.

6. What are the three most recently published papers that are relevant to this manuscript?

Risely, A., Müller-Klein, N., Schmid, D. W., Wilhelm, K., Clutton-Brock, T. H., Manser, M. B., & Sommer, S. (2023). Climate change drives loss of bacterial gut mutualists at the expense of host survival in wild meerkats. *Global Change Biology*, 29(20), 5816-5828.

Fontaine, S. S., & Kohl, K. D. (2023). Ectotherm heat tolerance and the microbiome: current understanding, future directions and potential applications. *Journal of Experimental Biology*, 226(12), jeb245761.

Fang, M., Lei, Z., Ruilin, M., Jing, W., & Leqiang, D. (2023). High temperature stress induced oxidative stress, gut inflammation and disordered metabolome and microbiome in tsinling lenok trout. *Ecotoxicology and environmental safety*, 266, 115607.

7. Provide the name and institutional email addresses of 5 potential reviewers.

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Thank you for considering our manuscript. We appreciate your time and look forward to your response.

Sincerely,

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